The 2D Location Test

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Below is a more concise English LaTex version. It focuses on the required inputs, physical model, inversion principle, and workflow rather than explaining every MATLAB field one by one.

1 Objective

This study develops a simple two-dimensional synthetic earthquake location framework based on P- and S-wave arrival times. The purpose is to examine and compare two location strategies:

- Absolute-arrival-time location, which uses the observed arrival time at each station;
- Differential-arrival-time location, which uses the arrival-time difference between two events recorded at the same station and for the same seismic phase.

The model assumes a homogeneous medium with constant P-wave and S-wave velocities. Earthquake events and seismic stations are randomly distributed in a two-dimensional domain. Synthetic arrival-time observations are then generated from the known event locations and origin times. Finally, the unknown event parameters are recovered by minimizing the misfit between predicted and observed arrival times.

2 Synthetic Model and Required Data

The synthetic model is defined by several basic physical and numerical parameters:

```
param.vp=6.0;
param.vs=param.vp/1.73;
param.Nsta=10;
param.Neve=30;
param.Domain=30;
param.nDim=2;
param.maxIter=500;
param.huberDelta=1.0;
param.reg=0.001;
```

The P-wave velocity is defined as:

$$v_P = 6.0 \text{ km/s},$$

and the S-wave velocity is defined by:

$$v_S = \frac{v_P}{1.73}.$$

The velocity ratio is selected to represent the common condition that S waves propagate more slowly than P waves. In this synthetic example, both velocities are assumed to be spatially constant. Therefore, the seismic medium is homogeneous and isotropic. This assumption simplifies the travel-time calculation and allows the location algorithm to be evaluated without the additional uncertainty caused by a complex velocity structure.

The parameter N_{sta} denotes the number of stations, while N_{eve} denotes the number of earthquake events. In the present example:

$$N_{\text{sta}} = 10, \qquad N_{\text{eve}} = 30.$$

The parameter `param.Domain` defines the size of the two-dimensional spatial domain. Both stations and events are randomly generated within:

$$0 \leq x \leq \text{Domain}, \qquad 0 \leq y \leq \text{Domain}.$$

For the current configuration:

$$0 \leq x, y \leq 30 \text{ km}.$$

The parameter `param.nDim=2` indicates that the inversion is performed in two dimensions. Thus, only the horizontal coordinates (x, y) are inverted. Although positions are stored in three-component form (x, y, z) for convenience, the third component is fixed as:

$$z = 0.$$

The parameter `param.maxIter` specifies the maximum number of iterations used by the optimizer. The parameter `param.huberDelta` controls the transition point of the Huber loss function, while `param.reg` controls the regularization strength applied to station timing corrections in the absolute-arrival inversion.

The synthetic data required by the location method can be summarized as follows:

- P-wave and S-wave velocities;
- station coordinates;
- event coordinates;
- event origin times;
- station-dependent timing biases;
- station- and phase-dependent biases;
- phase arrival times;
- phase labels, such as P or S;
- observation weights.

For real-data applications, station coordinates and phase arrival times should be obtained from an earthquake catalog or seismic phase-picking procedure. In the present work, these quantities are generated synthetically so that the true event locations are known and can be used to evaluate inversion performance.

3 Synthetic Arrival-Time Generation

The function `LOC_GenData(param)` generates the synthetic station distribution, event distribution, origin times, and phase arrival-time observations.

For each station s, the station coordinate is denoted by:

$$\mathbf{x}_s = (x_s, y_s, 0).$$

For each event e, the event coordinate is denoted by:

$$\mathbf{x}_e = (x_e, y_e, 0),$$

and the event origin time is denoted by:

$$t_{0,e}.$$

The travel distance between event e and station s is:

$$d_{e,s} = \left\| \mathbf{x}_e - \mathbf{x}_s \right\|.$$

For a two-dimensional model, this distance is calculated as:

$$d_{e,s} = \sqrt{(x_e - x_s)^2 + (y_e - y_s)^2}.$$

The theoretical travel time for phase p is:

$$\tau_{e,s,p} = \frac{d_{e,s}}{v_p},$$

where:

$$v_p = \begin{cases} v_P, & \text{for P wave,} \\ v_S, & \text{for S wave.} \end{cases}$$

The synthetic observed arrival time is constructed as:

$$T^{\text{obs}}_{e,s,p} = t_{0,e} + \tau_{e,s,p} + \delta t_s + \delta t_{s,p}.$$

Here, δt_s is the station-dependent arrival-time bias, and $\delta t_{s,p}$ is the station- and phase-dependent bias.

The station-dependent bias represents a timing error that affects all phases recorded by the same station. It may represent station clock uncertainty, timing synchronization error, or a station correction caused by local geological conditions. In the code, this term is generated as:

```
data.Dtsta=(rand(param.Nsta,1)-0.5)*disterp;
```

The station–phase bias represents an additional phase-dependent delay. For example, the P-wave and S-wave arrivals recorded at the same station may have different systematic biases because of phase picking uncertainty, local wave propagation effects, or simplified velocity assumptions. It is generated as:

```
data.Dtpha=(rand(param.Nsta,2)-0.5)*disterp;
```

The first column of `Dtpha` corresponds to P-wave bias, and the second column corresponds to S-wave bias.

In this model, the bias amplitude is controlled by:

$$\mathrm{disterp} = 0.8.$$

Therefore, the station and phase biases are sampled approximately within:

$$-0.4 \text{ s} \le \delta t \le 0.4 \text{ s}.$$

For each event-station pair, two observations are generated: one P-wave arrival and one S-wave arrival. Therefore, the total number of absolute arrival-time observations is:

$$N_{\mathrm{obs}} = N_{\mathrm{eve}} \times N_{\mathrm{sta}} \times 2.$$

For the present example:

$$N_{\mathrm{obs}} = 30 \times 10 \times 2 = 600.$$

Each observation must contain four essential identifiers or values:

- event index;
- station index;
- phase type, P or S;
- observed phase arrival time.

An observation weight is also assigned to each phase. In the current simulation:

$$w_P = 1.0, \qquad w_S = 0.6.$$

This means that P-wave arrival times are considered more reliable than S-wave arrival times. Such a weighting scheme is common because P-wave onsets are often clearer and easier to pick than S-wave onsets. The weights can be modified according to phase-picking quality, signal-to-noise ratio, or manual quality control in real applications.

4 Absolute-Arrival-Time Location

Absolute-arrival-time location uses the complete observed arrival time:

$$T^{\mathrm{obs}}_{e,s,p}.$$

The unknown parameters include the event locations, event origin times, and station timing corrections. For a two-dimensional problem, the absolute-location model vector can be expressed as:

$$\mathbf{m}_{\mathrm{abs}} = \left[x_1, y_1, \ldots, x_{N_{\mathrm{eve}}}, y_{N_{\mathrm{eve}}}, t_{0,1}, \ldots, t_{0,N_{\mathrm{eve}}}, \delta t_1, \ldots, \delta t_{N_{\mathrm{sta}}}\right]^{\mathrm{T}}.$$

The predicted arrival time is:

$$T^{\mathrm{pred}}_{e,s,p} = t_{0,e} + \frac{\|\mathbf{x}_e - \mathbf{x}_s\|}{v_p} + \delta t_s.$$

The station–phase bias $\delta t_{s,p}$ is included when synthetic observations are generated, but it is not explicitly included as an inversion parameter in the current absolute-location model. Therefore, even when the true event locations, origin times, and station biases are used, the absolute arrival-time residual may not be exactly zero.

The residual for the kth observation is defined as:

$$r_k = T_k^{\text{pred}} - T_k^{\text{obs}}.$$

The inversion minimizes a weighted robust objective function:

$$J_{\text{abs}} = \frac{1}{N_{\text{obs}}} \sum_{k=1}^{N_{\text{obs}}} w_k \rho_\delta(r_k) + \lambda \frac{1}{N_{\text{sta}}} \sum_{s=1}^{N_{\text{sta}}} |\delta t_s|.$$

The first term measures the mismatch between predicted and observed arrival times. The second term is a regularization term for station timing corrections.

The Huber loss function is defined as:

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}\frac{r^2}{\delta}, & |r| \le \delta, \\ |r| - \frac{1}{2}\delta, & |r| > \delta. \end{cases}$$

For small residuals, the Huber loss behaves similarly to least-squares inversion. For large residuals, it behaves approximately as an absolute-value loss function. As a result, it is less sensitive to outliers than the conventional squared-error objective function.

The regularization parameter λ, represented by `param.reg`, is introduced because origin times and station timing corrections are coupled. For any constant C:

$$t'_{0,e} = t_{0,e} + C,$$

$$\delta t'_s = \delta t_s - C,$$

the sum remains unchanged:

$$t'_{0,e} + \delta t'_s = t_{0,e} + \delta t_s.$$

Consequently, the predicted arrival time remains unchanged. This indicates that the absolute timing reference is not fully determined without an additional constraint. The regularization term suppresses excessively large station corrections and improves numerical stability.

The absolute-location inversion is initialized from a random model and then optimized iteratively. After optimization, the estimated event locations, origin times, and station corrections are stored in `NewAbsModel`.

5 Differential-Arrival-Time Location

The relative-location procedure is based on differential arrival times. For two events e_1 and e_2, recorded at the same station s with the same phase p, the differential time is:

$$\Delta T_{e_1,e_2,s,p}^{\text{obs}} = T_{e_1,s,p}^{\text{obs}} - T_{e_2,s,p}^{\text{obs}}.$$

Substituting the absolute arrival-time equation gives:

$$\Delta T_{e_1,e_2,s,p}^{\text{obs}} = \left(t_{0,e_1} + \frac{\|\mathbf{x}_{e_1} - \mathbf{x}_s\|}{v_p} + \delta t_s + \delta t_{s,p} \right) - \left(t_{0,e_2} + \frac{\|\mathbf{x}_{e_2} - \mathbf{x}_s\|}{v_p} + \delta t_s + \delta t_{s,p} \right).$$

After simplification:

$$\Delta T^{\mathrm{obs}}_{e_1,e_2,s,p} = (t_{0,e_1} - t_{0,e_2}) + \frac{\|\mathbf{x}_{e_1} - \mathbf{x}_s\|}{v_p} - \frac{\|\mathbf{x}_{e_2} - \mathbf{x}_s\|}{v_p}.$$

The station bias cancels:

$$\delta t_s - \delta t_s = 0,$$

and the station–phase bias also cancels:

$$\delta t_{s,p} - \delta t_{s,p} = 0.$$

This is the main advantage of differential arrival times. They reduce the influence of station-related common timing errors and emphasize the relative travel-time difference between events.

For N_{eve} events, the number of unique event pairs is:

$$\frac{N_{\mathrm{eve}}(N_{\mathrm{eve}} - 1)}{2}.$$

Each event pair is evaluated for every station and for both P and S phases. Therefore, the total number of differential-time observations is:

$$N_{\mathrm{diff}} = \frac{N_{\mathrm{eve}}(N_{\mathrm{eve}} - 1)}{2} \times N_{\mathrm{sta}} \times 2.$$

The unknown model parameters for differential location are:

$$\mathbf{m}_{\mathrm{diff}} = \left[x_1, y_1, \ldots, x_{N_{\mathrm{eve}}}, y_{N_{\mathrm{eve}}}, t_{0,1}, \ldots, t_{0,N_{\mathrm{eve}}}\right]^{\mathrm{T}}.$$

Unlike absolute location, station timing corrections are not included because they cannot be recovered from same-station, same-phase differential times.

The predicted differential time is:

$$\Delta T^{\mathrm{pred}}_{e_1,e_2,s,p} = t_{0,e_1} + \frac{\|\mathbf{x}_{e_1} - \mathbf{x}_s\|}{v_p} - t_{0,e_2} - \frac{\|\mathbf{x}_{e_2} - \mathbf{x}_s\|}{v_p}.$$

The differential-time residual is:

$$r_k = \Delta T^{\mathrm{pred}}_k - \Delta T^{\mathrm{obs}}_k.$$

The corresponding objective function is:

$$J_{\mathrm{diff}} = \frac{1}{N_{\mathrm{diff}}} \sum_{k=1}^{N_{\mathrm{diff}}} w_k \rho_\delta(r_k).$$

Because the station-related bias terms are removed through differencing, the differential-time inversion should generally produce a smaller data misfit than the absolute-time inversion for this synthetic model.

6 Inversion Workflow and Result Evaluation

The absolute and differential inversions follow the same general optimization procedure:

1. Generate an initial model containing random event locations and origin times;

2. Calculate predicted arrival times or predicted differential arrival times;
3. Compute residuals between predicted and observed data;
4. Evaluate the weighted Huber objective function;
5. Update the model parameters using an iterative optimizer;
6. Stop when the maximum iteration number is reached or the objective function converges;
7. Compare the inverted event locations with the known synthetic locations.

The quality of the inversion can be evaluated using the mean squared error:

$$\mathrm{MSE} = \frac{1}{N}\sum_{k=1}^{N}\left(T_k^{\mathrm{pred}} - T_k^{\mathrm{obs}}\right)^2,$$

for absolute arrival times, or:

$$\mathrm{MSE}_{\mathrm{diff}} = \frac{1}{N_{\mathrm{diff}}}\sum_{k=1}^{N_{\mathrm{diff}}}\left(\Delta T_k^{\mathrm{pred}} - \Delta T_k^{\mathrm{obs}}\right)^2,$$

for differential arrival times.

The event location error for event e can be calculated as:

$$E_e = \left\|\mathbf{x}_e^{\mathrm{inv}} - \mathbf{x}_e^{\mathrm{true}}\right\|.$$

A smaller location error indicates that the inverted location is closer to the known synthetic location.

The synthetic example is designed to verify the correctness of the travel-time calculation, inversion formulation, and optimization process. In real seismic applications, several additional factors should be considered, including three-dimensional event depth, nonuniform velocity models, phase-picking uncertainty, incomplete station coverage, event-station geometry, and realistic observational noise.

A practical extension is to replace the synthetic station and event parameters with real station coordinates and phase-picking data while retaining the same absolute- and differential-time inversion framework.